Multimodal Fake vs Real Job Posting Classification

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Abstract

Fraudulent job postings on large recruitment platforms closely resemble genuine postings, turning manual moderation into a costly and unreliable filter. We study this problem on the Fake Job Postings dataset (~17,9k items; 4.8% fraud) and build an explainable multimodal classifier that unifies text, company-logo images, and structured metadata via robust late-fusion rules. Our system uses strong single-modality encoders (TF–IDF + linear models for text; VGG16, ResNet-50, MobileNetV2, AlexNet for images; simple MLPs for categorical features), trains with stratified splits to address class imbalance, and evaluates with ROC–AUC, PR–AUC, accuracy, F1, and confusion matrices. We compare several fusion strategies designed for missing or weak modalities, including (i) calibrated weighted logit averaging, (ii) feature-level fusion, (iii) rank-based reciprocal rank fusion, (iv)conditional model fusion. Across experiments, multimodal fusion consistently improves over the best single modality and remains stable when one source is noisy or absent. To make decisions auditable, we integrate XAI at token, pixel, and modality levels: SHAP for tabular/text features, Integrated Gradients for text saliency, and Grad-CAM for image evidence. The resulting workflow produces transparent predictions and actionable diagnostics for moderators, demonstrating a practical path to deployable, interpretable fraud detection in online recruitment.

Keywords

Multimodal learning, Online recruitment fraud, Late fusion, Probability calibration, Isotonic regression, Reciprocal rank fusion, Explainable AI, TF–IDF, VGG16, ResNet–50, MobileNetV2, AlexNet, Class imbalance, Stratified split, ROC–AUC, PR–AUC, Threshold tuning

1 Introduction

Online recruitment platforms operate under asymmetric risk: a tiny fraction of fraudulent listings can cause out-sized harm, yet most posts are real. Prior work on job-post fraud (and related spam/abuse detection) has typically modeled *one* modality—usually text—or, at best, combined *two* modalities (e.g., text+metadata or text+image), with explanations limited to token- or pixel-level saliency. These approaches struggle when signals are missing, weak, or conflicting, and they rarely address calibration for operational thresholds. In this paper we advance beyond that with a *tri-modal* system that unifies long-form text, company-logo images, and structured attributes, and we study both learned fusion (final decision head, feature-level fusion, conditional end-to-end fusion for missing modalities) and calibrated post-hoc fusion. The result is a detector designed to be accurate under severe class imbalance, *calibrated* for threshold selection, and *auditable* across modalities—so that human moderators can trust and scrutinize its decisions.

2 Important Definitions and Problem Statement

2.1 Key Definitions

Fake Job Posting A job advertisement intentionally created to mislead applicants. These postings often mimic the structure or branding of legitimate listings but are designed to extract sensitive personal information, solicit payments, or direct users to malicious websites. Their subtlety makes manual detection difficult, highlighting the need for data-driven solutions.

Multimodal Learning A learning paradigm that integrates multiple sources of information, such as text, structured tabular data, and images into a unified predictive framework. In this project, multimodality allows the system to capture linguistic cues, hiring pattern irregularities, and visual inconsistencies associated with fraudulent job listings.

TF-IDF (Term Frequency–Inverse Document Frequency) A classic text-representation technique that quantifies how important a word is relative to both its document and the overall corpus. It supports traditional machine learning models by emphasizing terms that regularly appear in fraudulent postings.

BERT Embeddings Dense contextual embeddings generated using a pretrained BERT model. By capturing subtle semantic relationships and contextual meaning across job descriptions, requirements, and profiles, these embeddings serve as a strong foundation for the textual component of the classifier.

Convolutional Neural Network (CNN) A deep neural architecture specialized for extracting features from visual data. CNNs identify patterns such as shapes, textures, and layout anomalies, making them suitable for identifying inconsistencies in company logos, job-post images, or other visual elements related to a listing.

2.2 Problem Statement

Online recruitment platforms face a significant and growing challenge: distinguishing legitimate job postings from fraudulent ones. With a dataset of nearly 18,000 postings of which only about 4.8% are fraudulent, a simple heuristic or rule based approach quickly

breaks down. Fraudulent listings often blend convincingly with genuine ones, and the imbalance between real and fake posts makes the minority class especially hard to detect.

This project aims to develop a multimodal classification system capable of accurately identifying fake job postings. The system draws on three complementary modalities:

- Unstructured text, capturing suspicious linguistic patterns or unrealistic role descriptions
- Structured categorical metadata, identifying atypical combinations of employment type, experience levels, education requirements, and industry markers
- Images, assessing potential visual inconsistencies such as questionable logos or formatting cues

By combining these sources of evidence, the goal is to create a robust model that improves fraud detection accuracy and contributes to safer online hiring ecosystems.

3 Overview of Proposed Approach

This work presents a unified multimodal learning framework for fraud detection in job postings by leveraging information derived from three heterogeneous sources: *textual descriptions*, *structured categorical metadata*, and *job posting images*. Each modality captures complementary aspects of a job listing; linguistic intent, platform-level attributes, and visual cues, offering distinct perspectives on potentially fraudulent content. By jointly modeling these modalities within a coherent classification pipeline, the system aims to improve robustness, interpretability, and overall classification performance beyond what any single modality can achieve. The approach integrates comprehensive exploratory analysis, modality-specific preprocessing, and a diverse set of unimodal and multimodal models, culminating in a flexible fusion architecture capable of adapting to incomplete or partially missing modalities.

The overall system follows a multi-stage multimodal fusion pipeline:

(1) **Modality-specific encoders** independently transform raw text, categorical variables, and images into dense semantic embeddings.
(2) **Embedding alignment and projection** place all modality representations into a compatible latent space.
(3) **Fusion module** combines the multimodal embeddings into a unified representation.
(4) **Downstream classifier** generates the final prediction from the fused vector.

The framework is designed to remain flexible and modular, allowing the integration of multiple fusion strategies such as feature-level fusion, late/post-hoc fusion, and conditional fusion mechanisms. We explored various different strategies to analyze how multimodal interactions influence model behavior and performance.

To enhance interpretability, the system incorporates an **explainability (XAI) layer**. Using gradient-based, attention-based, and attribution-based techniques, the model highlights modality-specific evidence that drives each prediction. These explanations help validate model reliability and provide insights into the interplay between textual, categorical, and visual cues.

Overall, the proposed multimodal framework is designed to balance flexibility, predictive strength, and interpretability. By combining complementary representations and providing explainability mechanisms for each modality, the system supports both accurate fraud detection and transparent reasoning behind model decisions. The overall pipeline is shown in Figure 1.

4 Technical Details of Proposed Approaches / Systems

This section presents the complete technical design of the multimodal fraud-detection system, including preprocessing pipelines, modality-specific encoders, fusion mechanisms, CNN backbone experiments, training protocols, and integrated explainability components.

4.1 Data Processing Pipeline

A unified preprocessing pipeline prepares text, structured metadata, and images for multimodal learning.

Text Processing:

- Combined textual fields (title, location, department, description, requirements, company_profile and benefits)
- Tokenized using the BERT-base-uncased tokenizer with a maximum sequence length of 256
- TF–IDF vectorization used for classical ML baselines (Naive Bayes, Logistic Regression, SVM)

Metadata Processing:

- Categorical variables processed using:
  - Label encoding for neural models, with 0 reserved for missing categories
  - One-hot encoding optionally applied for classical ML baselines (e.g., Logistic Regression, Naive Bayes)
- Salary information processed as follows:
  - Extracted `salary_min` and `salary_max` from text ranges
  - Computed `salary_mean` as the average of min and max
  - Created a binned categorical feature `salary_category` (1–5) and assigned 0 for missing values
  - Added binary flag `salary_missing` for listings with no salary information
  - Missing values for numeric features (`salary_min`, `salary_max`, `salary_mean`) filled with 0

Image Processing:

- Resized to 224×224
- Normalized to ImageNet mean and standard deviation
- During multimodal fusion missing or invalid images were handled by substituting zero-tensor placeholders or 0.5(neutral) probabilities.

4.2 CNN Backbone Experiments

To establish strong image-only baselines and evaluate the role of visual features:

- **VGG16:** Frozen backbone with light augmentation; consistent ROC–AUC performance
- **MobileNetV2:** Lightweight and efficient; high ROC–AUC, suitable for constrained deployment

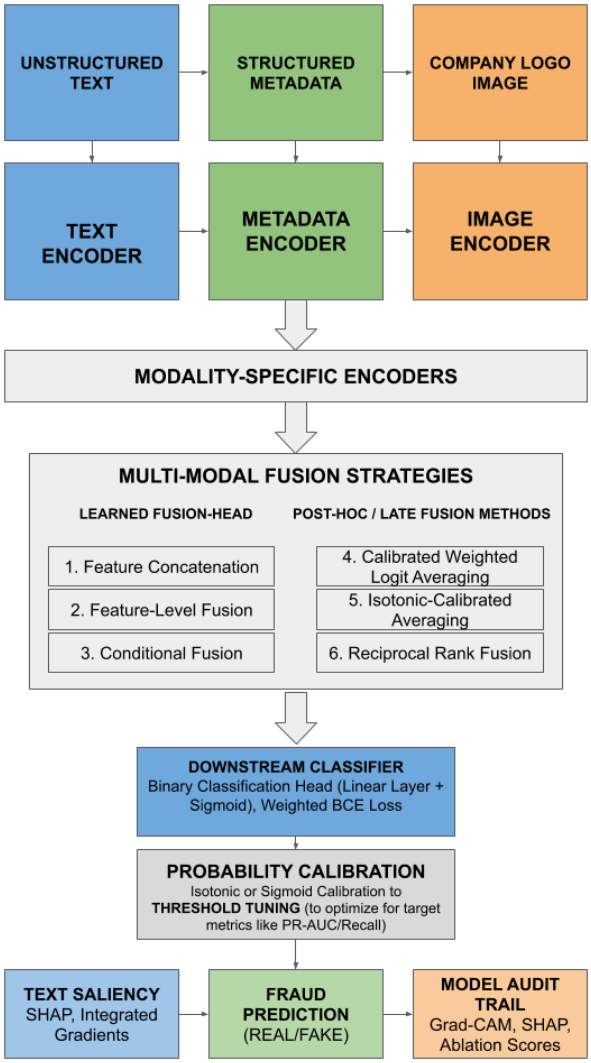


Figure 1: System architecture: modality encoders, fusion strategies, classifier head, and XAI outputs.

- **ResNet50:** Fine-tuned last convolutional blocks; strongest image-based generalization, chosen as primary image encoder
- **AlexNet:** Historical baseline; lower performance but informative for capacity comparison

4.3 Modality-Specific Encoders

Text Encoder:

- BERT-base-uncased transformer used to encode free-text fields
- CLS (pooler) token embedding serves as the 768-dimensional text representation

Categorical Encoder:

- Categorical features embedded via trainable embedding layers
- Embedded vectors projected to a 64-dimensional dense representation through an MLP: $4 \longrightarrow 64 \longrightarrow 64$

Image Encoder:

- Convolutional backbone (ResNet50, AlexNet) extracts deep visual features
- Global average pooling yields a 2048-dimensional fixed-length image embedding

4.4 Multimodal Fusion Strategies

4.4.1 Learned Fusion-Head Strategies.

(1) **Feature Concatenation:**
- Combines text and image embeddings by simple concatenation.
- Provides a straightforward way to jointly leverage information from different modalities.

- Missing modalities (e.g., absent images) can be handled using placeholders, allowing the model to continue using available features.
- Treats each modality independently before combining them in a single representation for downstream prediction.

(2) **Feature-Level Fusion:**
- Integrates all available embeddings (text, image, structured metadata) into a single compact representation.
- Allows the model to learn interactions between modalities at the feature level.
- Useful when multiple modalities provide complementary information that can jointly improve prediction quality.
- Projects heterogeneous embeddings into a shared space where cross-modal interactions can be captured.

(3) **Conditional Fusion:**
- Performs fusion of text, image, and structured embeddings conditionally based on which modalities are present.
- Missing modalities are dynamically masked to prevent the model from relying on unavailable data.
- Supports selective updating of the fusion head while keeping the modality encoders fixed, enabling robust learning even with partially missing data.
- Emphasizes flexible, context-aware integration of modalities rather than treating all inputs uniformly.

4.4.2 *Post-hoc / Late Fusion Methods.*

(1) **Calibrated Weighted Logit Averaging (C–WLA):** Calibrate per-modality probabilities on validation; convert to log-odds and combine with non-negative weights tuned to maximize PR–AUC.

(2) **Isotonic-Calibrated Averaging:** Calibrate each modality with isotonic regression; average calibrated probabilities; threshold chosen on validation for target metric.

(3) **Reciprocal Rank Fusion (RRF):** Convert modality scores to ranks; combine via $\mathrm{RRF}(x) = \sum_m \frac{1}{k+\mathrm{rank}_m(x)}$, robust to poorly calibrated scores.

4.5 Integrated Explainability (XAI)

Explainability is incorporated directly into evaluation for each modality.

SHAP Explanations:
- Text: token-level attribution via masking-based SHAP
- Image: inpainting-based SHAP highlighting important visual regions

Integrated Gradients:
- Gradients computed with respect to token embeddings
- Zero-embedding baseline to quantify contribution

Grad-CAM:
- Applied to the final convolutional layer of ResNet50
- Highlights discriminative visual patterns (e.g., logos, watermarks)

5 Experiments

5.1 Checkpoint 1: Text and Baseline Models

- **Baseline Models:**
  - **Complement Naive Bayes (TF-IDF + One-Hot Encoding):** Moderate ROC–AUC, interpretable via token contributions.
  - **Lasso Regression:** Slightly better ROC–AUC, strong feature selection properties.
  - **BERT Text Classifier (bert-base-uncased):** Achieved state-of-the-art ROC–AUC (≈ 0.99), serving as the primary text-only benchmark. Label encoding used for categorical features if included.

These experiments established that textual features are highly predictive and set the foundation for multimodal integration.

5.2 Checkpoint 2: Text+Metadata and CNN Image Experiments

- **Text + Categorical/Metadata Fusion Experiments:**
  - Fused text features from both BERT (bert-base-uncased) and Complement Naive Bayes with structured categorical features.
  - Encoding strategy:
    * BERT: Label encoding for categorical features
    * Complement Naive Bayes: One-hot encoding for categorical features
  - Trained a multimodal classifier using concatenated text and categorical embeddings.
  - Observed that combining text and metadata improves prediction over text-only baselines, especially for listings where structured information is discriminative.
- **CNN Image Models:**
  - **VGG16:** Fast convergence, moderate PR–AUC, but heavier than MobileNetV2.
  - **MobileNetV2:** Lightweight model suitable for deployment, moderate performance.
  - **ResNet50:** Best overall CNN performance, selected as final image encoder for fusion experiments.
  - **AlexNet:** Second-best performance, used in one of the feature-level fusion techniques.

5.3 Final Multimodal Fusion Experiments

We evaluated the three learned fusion-head and three strategies as described in Section 4.3.

5.3.1 *Fusion Strategy 1 – Feature Concatenation.*

- **Inputs:** Precomputed text embeddings (768-d from BERT) and image embeddings (2048-d from ResNet50).
- **Dataset:** Custom FusionDataset that includes missing image masks. Missing images replaced with zero-tensors.
- **Architecture:**
  - Linear (text+image) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear $\rightarrow$ Sigmoid
- **Loss / Optimizer:** Weighted BCE; Adam, LR=1e−3.
- **Training:** Mini-batch gradient descent with stratified train-validation split; early stopping based on ROC-AUC.
- **Observations:** Text features dominate predictions early in training; images improve classification for listings with logos or high-quality visuals.

Table 1: Performance Comparison of Baseline Models (Checkpoint 1)

Model Name	Accuracy	ROC-AUC	Precision	Recall
Transformer-Based Text Classifier (BERT)	0.99	0.99	0.99	0.99
Lasso Regression	0.95	0.95	0.97	0.95
Complement Naive Bayes	0.96	0.92	0.97	0.99

Table 2: CNN Backbone Performance (image models for Checkpoint 2)

Model Name	Accuracy	PR-AUC	Precision	Recall
ResNet	0.917	0.733	0.65	0.66
MobileNet	0.952	0.279	0.625	0.058
VGGNet	0.957	0.326	0.806	0.145
AlexNet	0.765	0.660	0.58	0.52

5.3.2 Fusion Strategy 2 – Feature-Level Fusion. We tried 2 different CNNs, Resnet50 and Alexnet, for this approach.

- **Inputs:** Concatenated embeddings of text, image, and structured metadata.
- **Architecture:** Feed-forward network projecting fused features to a hidden layer (128-d) followed by a single output logit.
- **Loss / Optimizer:** Weighted BCEWithLogitsLoss; Adam, LR=1e−3.
- **Training:** 10 epochs, validation at each epoch, metrics: ROC-AUC, accuracy, precision, recall, F1.
- **Observations:** Integrating all modalities jointly improves performance and robustness over unimodal baselines. Metadata features provide consistent secondary signals.

5.3.3 Fusion Strategy 3 – Conditional Fusion.

- **Inputs:** Text tokens and attention masks (BERT), image tensors (ResNet50), structured metadata features.
- **Dataset:** `MultiModalDataset` handles missing images using `has_img_mask`.
- **Architecture:**
  – Concatenate(text, image, struct) → Linear → ReLU → Dropout(0.3) → Linear → Sigmoid
  Only the fusion head is fine-tuned; encoders remain frozen.
- **Loss / Optimizer:** BCE; Adam, LR=1e−3 applied to fusion head parameters only.
- **Training:** Typically 3 epochs; supports robust handling of missing modalities.
- **Observations:** Maintains strong performance even with missing modalities; best used when images are partially missing.

5.3.4 Post-hoc / Late Fusion .

- **Methods:** C–WLA, Isotonic-calibrated averaging, Reciprocal Rank Fusion (RRF).
- Operate on modality *scores* without retraining encoders.
- Handles missing modalities gracefully by injecting neutral prior (0.5 probability) or mid-rank.
- **Observations:** Fused post-hoc scores slightly improve PR-AUC over single-modality models and provide robustness when certain modalities are weak or missing.

A quantitative comparison across the three learned fusion strategies is shown in Table 3. Conditional fusion achieves the strongest overall performance, with the highest ROC-AUC (0.9872), accuracy (0.9880), and F1 score (0.8716), while feature-level fusion benefits from higher recall.

5.4 Explainability Experiments

To make the multimodal fraud detector auditable and actionable, we integrated a compact explainability pipeline operating at three levels: token-level (text), pixel-level (image), and modality-level (ablation/contribution). The implementation (PyTorch + HuggingFace + SHAP + Captum) generates visualizations and JSON metadata for each validation sample; outputs are saved to the `xai_outputs` directory for manual inspection and downstream UI integration.

5.5 Methods and implementation details

The multimodal model is interpreted using five families of explainability methods:

- **Token-level SHAP.** A text-based SHAP explainer is used to estimate the contribution of each token. A small background set of training texts is used for sampling. For each validation example, SHAP masks parts of the input text and observes how the predicted fraud probability changes, producing a token-importance heatmap.
- **Integrated Gradients (IG).** IG is applied to the transformer input embeddings. A zero embedding is used as the baseline. The method computes a path integral of gradients to quantify the influence of each token on the final prediction. IG provides a complementary perspective to SHAP and helps confirm whether highlighted tokens truly influence the model output.
- **Image SHAP.** An image masker with inpainting is used to perturb different regions of the image. By measuring how predictions change when regions are removed, SHAP assigns

Table 3: Comparison of Multimodal Fusion Strategies (Validation Metrics)

Fusion Strategy	ROC-AUC	Accuracy	Precision	Recall	F1 Score
Conditional Fusion	0.9872	0.9880	0.9012	0.8439	0.8716
Calibrated Weighted Logit Averaging	0.9988	0.9935	0.9744	0.3869	0.9339
Isotonic-calibrated Averaging	0.9989	0.9937	0.9717	0.2557	0.9357
Reciprocal Rank Fusion	0.9761	0.9506	0.5350	0.0574	0.6408
Feature-Level Fusion (ResNet50)	0.9821	0.9740	0.6754	0.8902	0.7681
Feature-Head Concatenation	0.9834	0.9852	0.8659	0.8208	0.8427
Feature-Level Fusion (AlexNet)	0.9890	0.9809	0.7488	0.9133	0.8229

Figure 2: Grad-CAM visualization for a representative validation sample. The heatmap (overlaid on the original image) highlights spatial regions the ResNet-50 backbone attends to when predicting fraud. This output supports case-level visual inspection and complements token-level attributions.

importance values across the spatial grid. The resulting explanation highlights areas of logos or other visual elements that affect the model.

- **Grad-CAM.** Grad-CAM produces a class-specific activation heatmap using gradients flowing through the final convolutional layer of the ResNet–50 image encoder. The heatmap is upsampled and overlaid on the original image, revealing the regions that most strongly support a fraud or non-fraud prediction.
- **Modality-level ablation.** To quantify the contribution of each modality, we compute the change in prediction when one modality is replaced with a neutral placeholder. Formally, for modality m:

$$\Delta_m = P(\text{fraud} \mid \text{all}) - P(\text{fraud} \mid \text{without } m).$$

  A larger Δ_m indicates that modality m plays a stronger role in the final decision.
- **Faithfulness checks.** To assess causal validity, we remove the top-k most influential tokens and recompute the prediction. A substantial drop in predicted fraud probability indicates that the explanations are faithful to the model's internal reasoning.

5.5.1 Representative findings. Applying the XAI pipeline revealed repeatable, interpretable patterns:

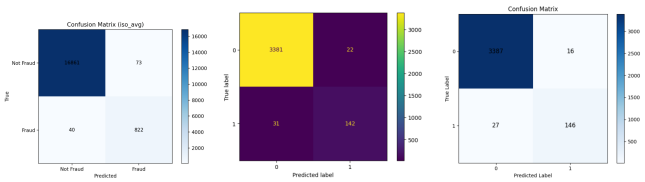
- **Text signals dominate.** SHAP and IG consistently highlight vague wording, urgency cues, missing benefits, and personal-information requests as fraud indicators.
- **Image cues.** Grad-CAM and image-SHAP often attend to logos and centered branding; missing, blurred, or noisy logos elevate fraud probability (higher `image_drop` when ablated).
- **Metadata role.** Binary flags (salary presence, company logo, telecommuting) provide reliable secondary signals and improve interpretability when combined with text and image evidence.
- **Faithfulness.** Token-removal tests (mask top-k tokens by IG/SHAP) produce meaningful probability decreases, supporting the causal relevance of the highlighted tokens.

5.5.2 Reproducibility notes and compute considerations.

- **SHAP cost.** Masking and inpainting operations are computationally expensive. Using 150 evaluations per sample (for text or image) results in many forward passes. A GPU is recommended; `max_evals` may be reduced when resources are limited.
- **Grad-CAM and IG.** Both methods require gradient computation, so `torch.no_grad()` must not be used during attribution. This increases memory use and compute time relative to standard inference.
- **Missing images.** When an image is unavailable, the pipeline falls back to a neutral placeholder (a zero tensor). This behavior should be noted when interpreting image-only analyses or Grad-CAM visualizations.
- **Saved artifacts.** All explanation artifacts (JSON files and PNG images) are written to the `xai_outputs/` directory, where they can be inspected manually or consumed by downstream tools.

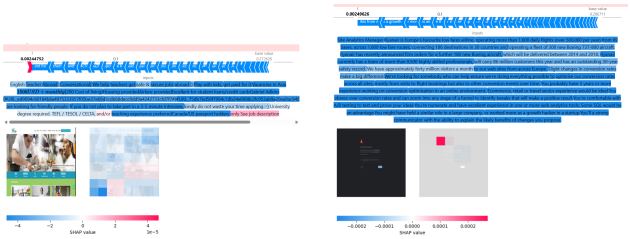
6 Results and Analysis

- **Strong performance of text-only models.** Transformer-based text classification (BERT) significantly outperformed traditional baseline models such as Lasso Regression and Complement Naive Bayes (Table 1). BERT achieved near-perfect performance across all metrics (Accuracy, ROC-AUC, Precision, Recall), highlighting that textual cues remain the most discriminative modality in fraudulent job postings.
- **Effectiveness of CNN backbones varies significantly.** Image-only experiments (Table 2) revealed large performance differences across CNN architectures. ResNet50 achieved the



(a) Post-hoc fusion model results on whole dataset **(b) Feature fusion results on validation dataset** **(c) Conditional fusion results on validation dataset**

Figure 3: Overview of three different fusion strategies evaluated in this work.



(a) Example 1 **(b) Example 2**

Figure 4: Sample explanations generated by our XAI module showing both text-based and image-based interpretation outputs.

highest overall accuracy and PR-AUC, confirming its suitability as an image encoder for downstream multimodal models. MobileNet and VGGNet showed competitive accuracy but extremely low recall, indicating that they miss a large portion of fraudulent cases. AlexNet performed the worst overall, confirming the importance of using modern deep feature extractors for visual signals.

- **Benefits of multimodal learning.** All multimodal fusion models surpassed single-modality image models and approached the strong text-only baselines, demonstrating that metadata and visual cues complement textual information. The inclusion of structured attributes stabilizes predictions when textual cues are ambiguous, while logos and posting images provide additional fraud signals.
- **Performance comparison of fusion strategies.** Table 3 shows that multiple fusion approaches achieved high accuracy and ROC-AUC, but differed substantially in their precision–recall trade-offs.
  - *Conditional fusion* achieved the best overall balance, with strong precision and recall and the highest robustness when images were missing or unreliable.
  - *Calibrated logit averaging* produced extremely high precision but very low recall, making it conservative in labeling fraud.
  - *Feature-level fusion* (ResNet50 and AlexNet variants) offered stable and interpretable representations, with AlexNet surprisingly outperforming ResNet50 in recall due to softer feature activations.
  - *Reciprocal Rank Fusion* performed poorly on recall, showing that ranking-based aggregation is less suitable for this classification task.

  Overall, fusion models show clear advantages over independent modality models, validating the importance of jointly modeling text, metadata, and images.
- **Insights from Explainable AI (XAI).** Model interpretability tools such as SHAP and Grad-CAM provided clear qualitative evidence supporting the fusion results:
  - Fraudulent postings consistently exhibited **vague, overly generic descriptions** and **sensational language**, which heavily influenced the text encoder.
  - Image-based explanations highlighted that **missing, low-quality, or stock logos** are strong indicators of fraud.
  - Structured metadata signals—such as **missing salary information**, **incomplete employer profiles**, and **anomalous posting dates**—frequently contributed to positive fraud predictions.

These patterns validate that each modality contributes unique and meaningful information to the overall fraud detection pipeline.

7 Related Work

Research on detecting fake job advertisements and fraudulent online recruitment has grown significantly. Below are a number of representative studies, organized by their methodological approaches and relevance to multimodal or machine-learning based detection.

7.1 Machine Learning and Traditional Approaches to Fake Job Detection

- Real or Fake Job Posting Detection Sridevi, Likitha, Chandana & Shamarthi (2024): This study uses a dataset of 17,880 entries and applies classical ML classifiers (K-Nearest Neighbors, SVM, Random Forest, Logistic Regression, Naïve Bayes, AdaBoost) over textual and structured features (job title, company profile, description, industry, salary, etc.). The authors report high classification performance, especially using Random Forest [6].
- Fake Job Detection and Analysis Using Machine Learning and Deep Learning Algorithms Nagarajan et al. (2025): This paper compares traditional ML and deep-learning models on cleaned and pre-processed job-post data. It emphasizes careful data cleaning, feature engineering (unigrams/bigrams), and shows that deep-learning models outperform classical approaches in detecting fake job posts [2].
- Online Fake Job Advertisement Detection using Machine Learning Techniques (2024): This work uses preprocessing, TF-IDF vectorization, and ensemble classifiers (e.g., Random Forest) to detect fraudulent job ads. It argues for scalability and automation as a defense against the growing volume of fraudulent postings [9].
- Predicting Fake Job Posts Using Machine Learning Models Boka (2024): This study applies a variety of classifiers (decision trees, logistic regression, SVM, naïve Bayes, random forest, multilayer perceptron) on a dataset of job ads, using

data mining algorithms like Apriori for feature selection before classification. The work demonstrates the feasibility of automated fake-job detection using ML techniques [3].

7.2 Deep Learning & Multimodal / Advanced Feature Approaches

- Detecting Fake Job Postings Using Bidirectional LSTM Pillai (2023): This paper proposes a Bi-LSTM model over textual and numeric features to distinguish fake from real job ads. The authors report a high performance: 98.71% accuracy and 0.91 ROC AUC, underscoring the potential of sequence models for semantic analysis of job descriptions [8].
- Detecting Online Recruitment Fraud through Deep Learning 2025 study using CNN-based feature extraction (CNN2D) combined with structured data to classify fraudulent job ads. This moves beyond simple text features to richer learned feature representations, improving classification robustness [10].
- Visualizing and Clustering Fake Job Postings (2025): This work explores visualization and clustering to uncover patterns common to fake job ads such as industry-targeting trends, role types, and suspicious textual patterns. Though not a classification model, it provides valuable qualitative insights that can guide feature engineering or data sampling strategies [4].

7.3 Insights from Related Domains: Phishing, Spam & Multimodal Detection

- Advanced BERT and CNN-Based Computational Model for Phishing Detection in Enterprise Systems Gupta et al. (2024): Although focused on phishing email detection rather than job ads, this work demonstrates the power of combining pretrained language models (BERT) for text feature extraction with CNN-based classification for phishing content detection. The conceptual similarity supports using hybrid textual + deep-learning models for scam detection tasks [5].
- Multimodal Framework for Phishing Attack Detection and Mitigation Murhej et al. (2025): This study integrates multiple data sources, SMS, email body, URLs, CSS, and images to detect phishing attacks. The use of multimodal inputs (text + structural + visual/URL features) offers a compelling precedent for applying such an approach to job-post fraud detection [7].
- Spam Detection Using Bidirectional Transformers and Machine Learning Guo et al. (2023): This research employs transformer-based embeddings (e.g., BERT) combined with classical classification methods to detect spam content. Though applied to spam rather than job ads, it demonstrates how advanced NLP embeddings improve detection of malicious or deceptive content, which is relevant for detecting job-ad fraud [1].

7.4 Summary of Observed Trends and Gaps

- Traditional machine-learning approaches (TF-IDF + classical classifiers) remain popular and often yield high accuracy, but they rely heavily on manual feature engineering.
- Deep-learning models, especially sequence models (e.g., Bi-LSTM) and CNN-based approaches, offer more capacity to learn nuanced semantic or structural patterns without manual feature crafting.
- Multimodal methods, though still more common in phishing or spam detection, show significant promise: they combine textual, structural (metadata), and potentially visual cues to capture complex fraud signals that single-modality models may miss.
- There remains a relative scarcity of research applying full multimodal systems (text + structured metadata + images) to fake job posting detection, representing a potential gap and opportunity for novel contributions.

8 Conclusion

We presented an explainable multimodal fraud detector for online job postings that integrates text, company-logo images, and structured metadata through lightweight late-fusion rules. The design addresses operational constraints encountered by platforms: rare positives, heterogeneous and partially missing inputs, and the need for decisions that are auditable by humans.

Empirically, transformer-based text models and a ResNet-50 image backbone provide strong single-modality baselines; calibrated late-fusion further improves robustness and precision–recall behavior, particularly under class imbalance. The interpretability layer (token-level attributions, Grad-CAM visualizations, and feature importances) offers clear, case-level diagnostics that align model outputs with actionable cues such as vague language, suspicious branding, or inconsistent metadata.

In practice, the system can operate as a triage layer: high-confidence legitimate posts pass automatically, high-risk cases are flagged with explanations, and uncertain cases are routed to human review. Future work includes: (i) end-to-end multimodal training with uncertainty estimation; (ii) cost-sensitive thresholding to prioritize recall on the minority class; (iii) broader signals (e.g., domain reputation and behavioral metadata); and (iv) stress tests for missing-modality and distribution shift. Overall, the proposed pipeline delivers a balanced combination of accuracy, robustness, and auditability suitable for real-world fraud prevention.

GitHub Repository Link

https://github.com/anagha-h/Multimodal-Fake-Job-Posting-Classification

References

[1] 2022. Journal of Computational and Cognitive Engineering.

[2] C S Anita, P Nagarajan, G Aditya Sairam, P Ganesh, and G Deepakkumar. 2021. Fake job detection and analysis using machine learning and deep learning algorithms. *Rev. Gest. Inov. Tecnol.* 11, 2 (June 2021), 642–650.

[3] Melkamu Boka. 2024. Predicting fake job posts using machine learning models. *SSRN Electron. J.* (2024).

[4] Chee Keong Ch'ng and Xiang Yi Wong. 2025. Visualizing and clustering fake job postings: Data-driven insights for fraud detection. *BAREKENG J. ILMU MAT. DAN TERAP.* 20, 1 (Nov. 2025), 0865–0880.

[5] Brij B Gupta, Akshat Gaurav, Varsha Arya, Razaz Waheeb Attar, Shavi Bansal, Ahmed Alhomoud, and Kwok Tai Chui. 2024. Advanced BERT and CNN-based computational model for phishing detection in enterprise systems. *Comput. Model. Eng. Sci.* 141, 3 (2024), 2165–2183.

[6] K. Sridevi, G. Likitha, P. Chandana, and Shrutika Shamarthi. 2024. Real or fake job posting detection. *Int Res J Adv Engg Mgt* 2, 08 (Aug. 2024), 2499–2504.

[7] Mahmoud Murhej and G Nallasivan. 2025. Multimodal framework for phishing attack detection and mitigation through behavior analysis using EM-BERT and SPCA-BASED EAI-SC-LSTM. *Front. Commun. Netw.* 6, 1587654 (July 2025).
[8] Aravind Sasidharan Pillai. 2023. Detecting fake job postings using bidirectional LSTM. (April 2023). arXiv:2304.02019 [cs.LG]
[9] Publication Fee. [n. d.]. Low processing fee journal in engineering : Impact factor: 8.699. https://www.ijirset.com. Accessed: 2025-12-4.
[10] Ms S Sai Jaswanthi, D Bright Anand, R Karunia Krishnapriya, Mr Pandreti, and T Senthil. [n. d.]. Detecting online recruitment fraud through deep learning using hybrid neural networks. https://www.ijcrt.org/papers/IJCRT25A4264.pdf. Accessed: 2025-12-4.